

# Scheme 5

## Psychological Concept Clustering and Framework Mapping Scheme

*Second-order normalization of detected psychological concepts*

**DRAFT FOR EXPERT EVALUATION.** These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes. The workflow itself has since been validated end to end in two cross-provider test runs, in which three large language models from three different providers applied the abstract-level and the full-text scheme independently and their agreement was quantified per coded field. The full run on the review corpus is deliberately held until this evaluation is complete.

### At a Glance

#### Workflow position

Post-detection concept normalization over Stage 2 and Stage 3 concept strings.

#### Operational mode

Fixed pattern-based concept detection upstream, followed by LLM clustering for higher-order normalization.

#### Unit of analysis

The set of unique concept strings across the corpus, not whole records.

#### Provenance basis

coding.py concept and framework patterns plus the clustering prompt.

#### Research questions

RQ3 (which psychological concepts and frameworks dominate)

#### Tagline

Fixed pattern detection upstream, LLM clustering for higher-order normalization

### Canonical Source Paths

- `src/03_pipeline/bps_review/extraction/coding.py`
- `src/01_protocol/codebooks/stage2_codebook.md`
- `src/01_protocol/codebooks/stage3_codebook.md`
- `src/05_data/interim/extraction/llm_concept_clusters.json`

### Purpose

This scheme standardizes higher-order concept mapping after concept detection. It groups extracted psychological concepts from chronic pain review records into interpretable families and links them to likely theoretical frameworks. The goal is cross-record comparability when the raw concepts are heterogeneous, overlapping, or variably named.

It is a second-order coding scheme: it does not classify whole records, it normalizes the concept vocabulary itself.

## Upstream Concept Detection Seeds (current)

---

*The 16 fixed patterns used to detect concepts upstream. The taxonomy below expands and structures these.*

- fear-avoidance, catastrophizing, kinesiophobia, depression, anxiety, coping, self-efficacy, acceptance
- illness perceptions, pain beliefs, distress, expectations, emotion regulation, mindfulness, trauma, sleep

## Upstream Framework Seeds (current)

---

*The 7 fixed framework patterns.*

- fear-avoidance model, cognitive behavioral therapy, acceptance and commitment therapy
- illness perception framework, operant learning, social cognitive theory, self-regulation

## Cluster Output Schema (current)

---

*The clustering LLM returns strict JSON with a top-level clusters list.*

### **clusters**

Top-level list of normalized concept clusters.

### **family**

Higher-order concept family label for the cluster.

### **members**

The original concept strings grouped into that family. (*free text*)

### **possible\_frameworks**

Likely theoretical or therapeutic frameworks associated with the family. (*free text*)

## Comprehensive Psychological Concept Taxonomy

---

*The comprehensive concept space that the clustering step should normalize toward. Each family is aligned one to one with a psychological subdomain of the Scheme 6 ontology, so the concept map and the semantic ontology share a single vocabulary. Members are representative constructs (not exhaustive); frameworks are the theories most often invoked for that family in chronic pain research. Experts are asked to add, move, rename, or remove members and frameworks.*

### **Catastrophizing and negative cognitive appraisal**

Exaggerated negative interpretation and mental amplification of pain and its threat value. *Ontology subdomain:* Catastrophizing and Negative Cognitive Appraisal. *Constructs:* pain catastrophizing, rumination, magnification, helplessness, negative appraisal, worry about pain. *Frameworks:* fear-avoidance model, cognitive-behavioral model, communal coping model.

### **Fear, avoidance and pain-related fear**

Fear of pain, movement, or reinjury and the avoidance or escape behaviour it motivates. *Ontology subdomain:* Fear Avoidance and Pain Related Fear. *Constructs:* kinesiophobia, fear of movement or reinjury, pain-related fear, avoidance behaviour, escape behaviour, threat hypervigilance. *Frameworks:* fear-avoidance model, avoidance-endurance model, operant learning.

### **Depression, low mood and negative affect**

Depressive symptoms and sustained negative affect linked to living with pain. *Ontology subdomain:* Depression Emotional Distress and Affect. *Constructs:* depression, depressive symptoms, anhedonia, hopelessness, demoralization, negative affect. *Frameworks:* cognitive model of depression, diathesis-stress, mutual maintenance model.

### **Anxiety and psychological reactivity**

Anxious apprehension and heightened physiological or cognitive reactivity to pain. *Ontology subdomain:* Anxiety and Psychological Reactivity. *Constructs:* anxiety, pain anxiety, health anxiety, anxiety sensitivity, physiological reactivity. *Frameworks:* anxiety sensitivity model, fear-avoidance model.

### **Self-efficacy, control and mastery**

Beliefs about ability to manage pain and exert control over outcomes. *Ontology subdomain:* Self Efficacy Control Beliefs and Perceived Mastery. *Constructs:* pain self-efficacy, perceived control, locus of control, mastery, agency, confidence. *Frameworks:* social cognitive theory, self-efficacy theory.

### **Acceptance, psychological flexibility and mindfulness**

Willingness to experience pain without avoidance while pursuing valued activity. *Ontology subdomain:* Acceptance Psychological Flexibility and Mindfulness. *Constructs:* pain acceptance, psychological flexibility, values-based action, committed action, present-moment awareness, cognitive defusion, willingness. *Frameworks:* acceptance and commitment therapy, psychological flexibility model, mindfulness-based approaches.

### **Coping strategies and adjustment**

Cognitive and behavioural efforts to manage pain and the resulting adjustment. *Ontology subdomain:* Pain Coping Strategies and Adjustment. *Constructs:* active coping, passive coping, problem-focused coping, emotion-focused coping, adaptive coping, maladaptive coping, adjustment. *Frameworks:* transactional model of stress and coping, self-regulation.

### **Attention, vigilance and pain processing**

How attention is captured by or directed away from pain and bodily threat. *Ontology subdomain:* Attention Vigilance and Pain Processing. *Constructs:* attentional bias, hypervigilance, distraction, pain interruption, attentional control, somatic focus. *Frameworks:* threat interpretation model, cognitive-affective model of interruption.

### **Illness beliefs, pain representations and meaning**

Beliefs about the nature, cause, timeline, and consequences of pain and its meaning. *Ontology subdomain:* Illness Beliefs Pain Representations and Meaning. *Constructs:* illness perceptions, pain beliefs, causal attributions, timeline beliefs, consequence beliefs, meaning of pain, illness identity. *Frameworks:* common-sense model of self-regulation, illness perception framework.

### **Cognitive-behavioural and psychotherapeutic approaches**

Structured psychological techniques targeting pain cognition and behaviour. *Ontology subdomain:* Cognitive Behavioral and Psychotherapeutic Approaches. *Constructs:* cognitive restructuring, behavioural activation, graded activity, graded exposure, relaxation, psychoeducation. *Frameworks:* cognitive-behavioral therapy, operant and respondent conditioning.

### **Third-wave, ACT and contextual approaches**

Contextual and acceptance-based therapies that target the function of experience. *Ontology subdomain:* Third Wave Therapies ACT and Contextual Approaches. *Constructs:* acceptance, defusion, values

clarification, self-as-context, mindfulness practice, functional contextualism. *Frameworks*: acceptance and commitment therapy, contextual behavioural science, mindfulness-based stress reduction.

### **Resilience, positive psychology and growth**

Protective strengths and positive adaptation despite persistent pain. *Ontology subdomain*: Resilience Positive Psychology and Post Traumatic Growth. *Constructs*: resilience, optimism, hope, benefit-finding, post-traumatic growth, positive affect, gratitude. *Frameworks*: broaden-and-build theory, resilience frameworks.

### **Identity, self-concept and pain biography**

How pain reshapes self-concept, roles, and life narrative. *Ontology subdomain*: Identity Self Concept and Chronic Pain Biography. *Constructs*: pain identity, self-discrepancy, biographical disruption, role loss, self-concept, loss of self. *Frameworks*: self-discrepancy theory, narrative and biographical approaches.

### **Trauma, adversity and life events**

Traumatic and adverse experiences that predispose to or maintain chronic pain. *Ontology subdomain*: Trauma Adverse Childhood and Life Events. *Constructs*: post-traumatic stress, trauma, adverse childhood experiences, abuse history, life stressors, victimization. *Frameworks*: mutual maintenance model, diathesis-stress, shared vulnerability model.

### **Personality and individual differences**

Stable traits and dispositions that shape pain experience and response. *Ontology subdomain*: Personality Psychological Traits and Individual Differences. *Constructs*: neuroticism, negative affectivity, harm avoidance, perfectionism, alexithymia, trait anxiety. *Frameworks*: trait vulnerability models, diathesis-stress models.

### **Cognitive function and executive processes**

Cognitive resources and executive processes affected by or affecting pain. *Ontology subdomain*: Cognitive Function Executive Processes and Brain Health. *Constructs*: attention, working memory, executive function, cognitive load, mental fatigue or brain fog, processing speed. *Frameworks*: cognitive resource models, interruption models.

### **Motivation, goal pursuit and engagement**

Motivational dynamics of pursuing valued goals alongside pain. *Ontology subdomain*: Motivational Processes Goal Pursuit and Engagement. *Constructs*: goal pursuit, goal conflict, approach and avoidance motivation, activity engagement, endurance, valued goals. *Frameworks*: self-regulation, motivational control theory, goal pursuit models.

### **Healthcare-seeking, adherence and engagement**

Behaviour around seeking, engaging with, and adhering to care. *Ontology subdomain*: Healthcare Seeking Treatment Adherence and Engagement. *Constructs*: treatment adherence, healthcare utilization, help-seeking, therapeutic alliance, treatment expectations, engagement. *Frameworks*: health belief model, common-sense model of self-regulation, expectancy models.

### **Emotion regulation and pain affect processing**

Strategies for regulating emotion and processing the affective dimension of pain. *Ontology subdomain*: Emotional Regulation and Pain Affect Processing. *Constructs*: emotion regulation, expressive suppression, cognitive reappraisal, alexithymia, emotional awareness, affect regulation. *Frameworks*: process model of emotion regulation, emotion regulation frameworks.

### **Mental health comorbidity and wellbeing**

Co-occurring mental health conditions and overall psychological wellbeing. *Ontology subdomain*:

Mental Health Comorbidity and Psychological Wellbeing. *Constructs:* comorbid depression or anxiety, distress, quality of life, wellbeing, suicidality, life satisfaction. *Frameworks:* mutual maintenance model, biopsychosocial wellbeing models.

### Analytic Role

---

- Preserves the original detected concepts while improving comparability across records.
- Supports synthesis questions about which psychological concepts and frameworks dominate the corpus.
- Allows concept families to be interpreted above raw string matching.

### Primary Outputs Using This Scheme

---

- `src/05_data/interim/extraction/llm_concept_clusters.json`